

require exact EOT solutions

convergence is often impractical



automatic differentiation

Limitations of Exact Derivatives

- Closed-form gradient expressions exist in both Eulerian and Lagrangian settings
- They **require exact EOT solutions** (dual potentials $\mathbf{f}$, $\mathbf{g}$ or primal coupling π)
- But we often compute these only approximately (e.g., fixed # of Sinkhorn iters)
- Running Sinkhorn to **convergence is often impractical** in large-scale training
- Computing/storing Jacobians can be expensive/intractable for complex models
- A scalable and flexible alternative: **automatic differentiation** of Sinkhorn

What Makes a Good ML Loss Function?

Optimizable

- Be differentiable (preferably smooth)
- Provide informative gradients for optimization
- Allow for principled regularization (e.g., via convexity or smoothness)

Practical & Scalable

- Scale well computationally (especially in deep learning)
- Work with mini-batching and SGD
- Integrate easily into modern ML pipelines

Aligned with the Learning Task

- Align well with the task's performance metric
- Be robust to noise, outliers, misspecification
- Reflect the structure of the underlying data domain